

Report

Prepared By

NIDA NOOR

PKCERT AI AND SOFTWARE DEVELOPMENT INTERNSHIP

Contents

1. Introduction	3
2. Dataset Description	3
Features	3
3. Data Preprocessing	3
4. Classification Models	4
Support Vector Machine (SVM)	4
Evaluation Metrics	4
Advantages	4
Limitations	4
k-Nearest Neighbors (kNN)	5
Evaluation Metrics	5
Advantages	5
Limitations	5
5. Performance Comparison	5
6. Results	6
Support Vector Machine	6
k-Nearest Neighbors	7
7. Recommendation	8
8. Conclusion	8

TASK 09

Support Vector Machine (SVM) & k-Nearest Neighbors (kNN)

1. Introduction

The objective of this project is to implement and compare two supervised machine learning classification algorithms, **Support Vector Machine (SVM)** and **k-Nearest Neighbors (kNN)**, using the **Diabetes Dataset**. Both models were trained to predict whether a patient has diabetes based on several medical attributes. Their performance was evaluated using standard classification metrics to determine the most suitable model for the dataset.

2. Dataset Description

The project uses the **diabetes.csv** dataset.

Features

Feature	Description
Pregnancies	Number of pregnancies
Glucose	Plasma glucose concentration
BloodPressure	Diastolic blood pressure
SkinThickness	Skin fold thickness
Insulin	Serum insulin
BMI	Body Mass Index
DiabetesPedigreeFunction	Genetic diabetes likelihood
Age	Age of the patient
Outcome	Target variable (0 = Non-Diabetic, 1 = Diabetic)

3. Data Preprocessing

The following preprocessing steps were performed:

- Loaded the dataset using Pandas.
- Explored the dataset structure and statistics.
- Replaced invalid zero values with missing values (NaN).
- Filled missing values using the median.
- Separated features (X) and target variable (y).
- Applied StandardScaler to normalize numerical features.
- Split the dataset into 80% training data and 20% testing data.

4. Classification Models

Support Vector Machine (SVM)

Support Vector Machine is a supervised learning algorithm that classifies data by finding the optimal hyperplane that maximizes the margin between different classes. The Radial Basis Function (RBF) kernel was selected because it effectively handles non-linear relationships within the dataset.

Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

Advantages

- Effective for high-dimensional datasets.
- Handles non-linear classification using kernel functions.
- Good generalization capability.
- Less prone to overfitting.

Limitations

- Computationally expensive for large datasets.
- Sensitive to kernel and parameter selection.

- Less interpretable than simpler models.

k-Nearest Neighbors (kNN)

k-Nearest Neighbors is a supervised learning algorithm that classifies new observations based on the majority class among their nearest neighbors. In this project, **K = 5** was selected to balance bias and variance.

Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

Advantages

- Simple and easy to implement.
- No explicit training phase.
- Effective for small datasets.
- Can model complex decision boundaries.

Limitations

- Slow prediction on large datasets.
- Sensitive to feature scaling.
- Performance depends on selecting an appropriate K value

5. Performance Comparison

Metric	SVM	kNN
Accuracy	74.68%	73.38%
Precision	66.67%	61.29%
Recall	58.18%	69.09%
F1-Score	62.14%	64.96%

6. Results

Both models performed well on the diabetes dataset.

Support Vector Machine

- Accuracy: **74.68%**
- Precision: **66.67%**
- Recall: **58.18%**
- F1-Score: **62.14%**

SVM MODEL EVALUATION

Accuracy : 0.7468

Precision: 0.6667

Recall : 0.5818

F1-Score : 0.6214

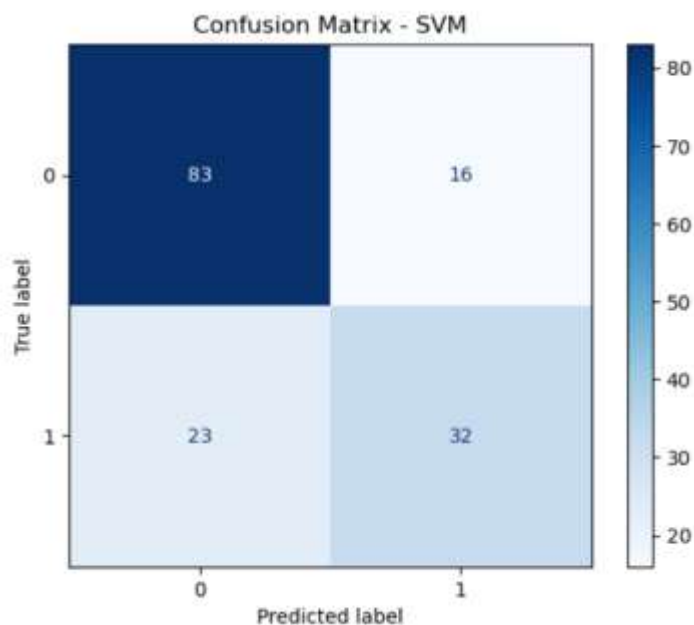
Confusion Matrix:

[[83 16]

[23 32]]

Classification Report:

	precision	recall	f1-score	support
0	0.78	0.84	0.81	99
1	0.67	0.58	0.62	55
accuracy			0.75	154
macro avg	0.72	0.71	0.72	154
weighted avg	0.74	0.75	0.74	154



k-Nearest Neighbors

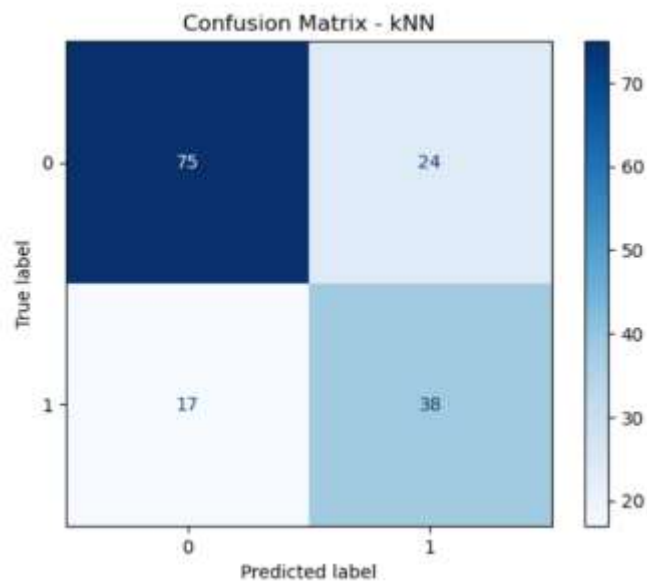
- Accuracy: **73.38%**
- Precision: **61.29%**
- Recall: **69.09%**
- F1-Score: **64.96%**

kNN MODEL EVALUATION
Accuracy : 0.7338
Precision: 0.6129
Recall : 0.6909
F1-Score : 0.6496

Confusion Matrix:
[[75 24]
[17 38]]

Classification Report:

	precision	recall	f1-score	support
0	0.82	0.76	0.79	99
1	0.61	0.69	0.65	55
accuracy			0.73	154
macro avg	0.71	0.72	0.72	154
weighted avg	0.74	0.73	0.74	154



SVM achieved the highest **Accuracy** and **Precision**, whereas kNN achieved higher **Recall** and **F1-Score**.

7. Recommendation

Although both algorithms produced competitive results, **Support Vector Machine (SVM)** is recommended for this dataset because it achieved the highest overall accuracy and precision, indicating better overall classification performance.

However, if the primary objective is to identify as many diabetic patients as possible and minimize false negatives, **kNN** may be preferred due to its higher recall.

8. Conclusion

This project successfully implemented and evaluated two supervised machine learning classification algorithms using the Diabetes Dataset. After preprocessing the data and evaluating both models, SVM demonstrated better overall classification accuracy, while kNN showed superior recall. The comparison highlights that selecting an appropriate model depends on the application's objectives. For balanced overall performance, SVM is recommended for diabetes prediction on this dataset.